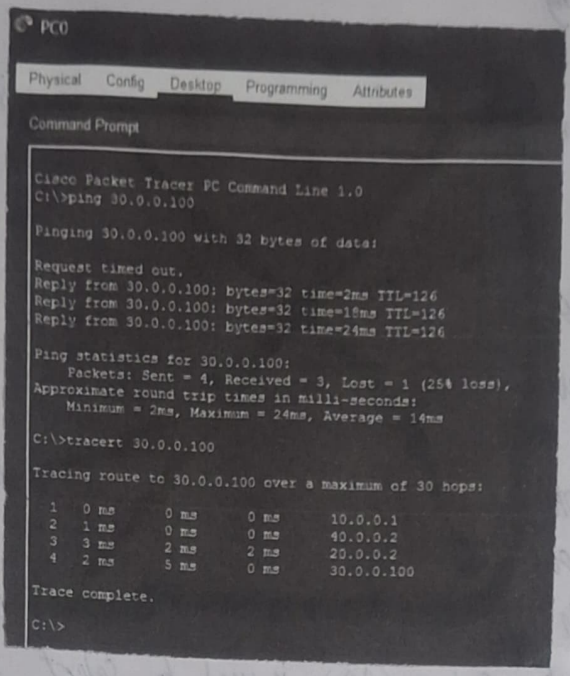
11a) Static Routing

Aim:- To simulate and configure static routing using Cisco Packet Tracer

Observation:-

1.) Static routing involves manually adding routes to a routers routing table
2.) Routers automatically knows the network directly connected to the interfaces
3.) Routes are manually added for networks that are available through other routers
4.) Both main and backup routes can be configured for a destination
5.) Administrative Distance (AD) is used to select the preferred static route.
6.) The routing table is check using:
Show ip route static
7.) Connectivity and routing paths are verified using ping and traceroute
8.) When the main route fails, the back up route can be used for forwarding packets




```
C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  1    0 ms       0 ms       0 ms       10.0.0.1
  2    3 ms       1 ms       1 ms       20.0.0.2
  3    0 ms       0 ms       1 ms       30.0.0.100

Trace complete.
```

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/1/1
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down
```

```
Router(config-if)#interface serial 0/1/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
```

```
C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    1 ms      7 ms      4 ms      40.0.0.2
  3    9 ms      7 ms      1 ms      20.0.0.2
  4    2 ms      0 ms      4 ms      30.0.0.100

Trace complete.

C:\>
```

Result :-

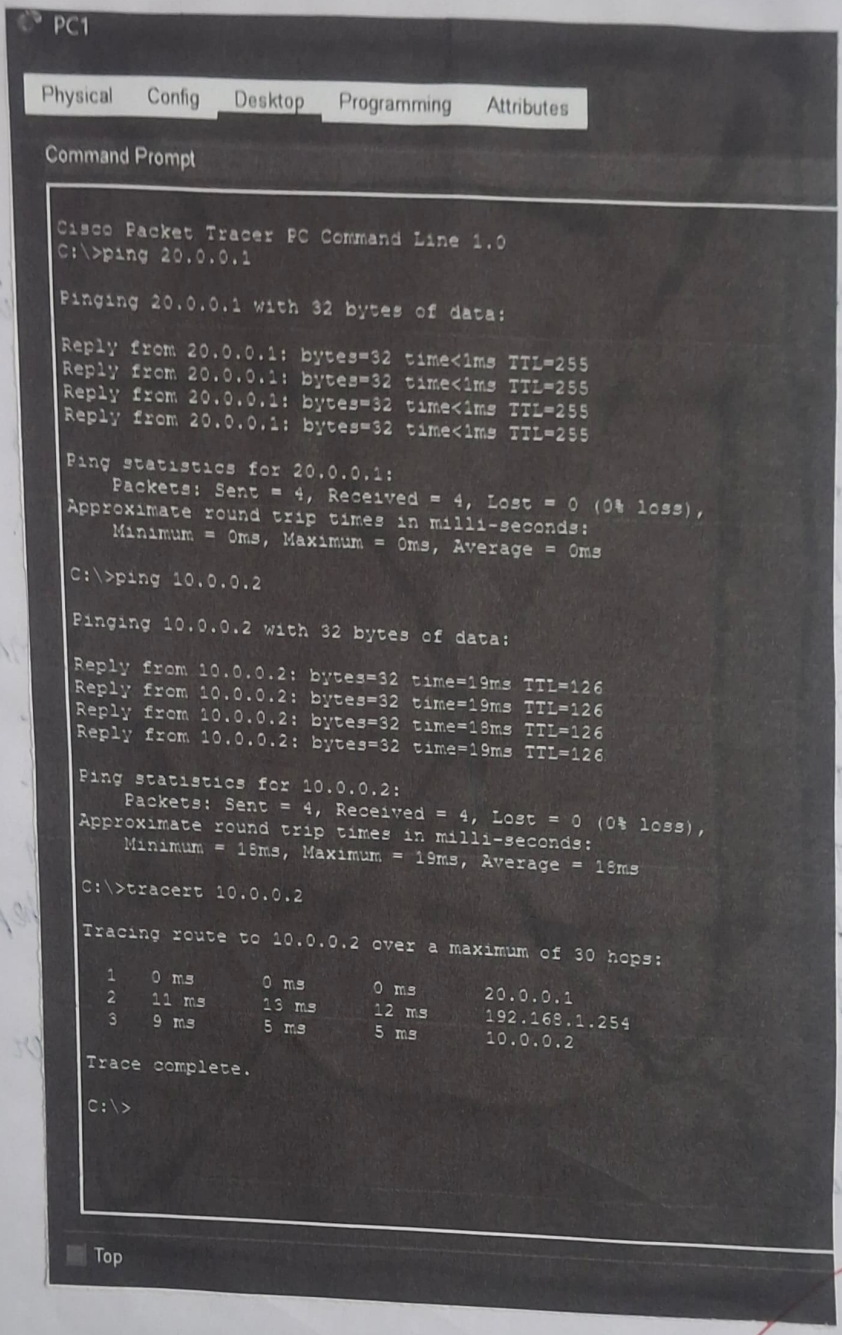
Thus, static routes were successfully configured and verified, including the use of backup route when the main route becomes unavailable

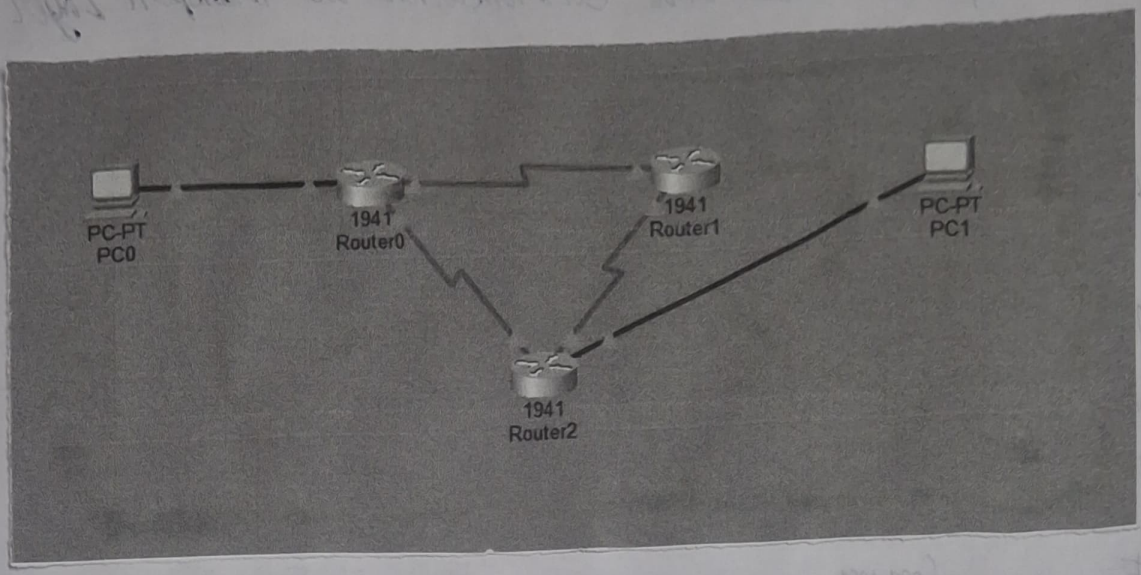
11.) b) RIP - Dynamic Routing

Aim :- To simulate and configure RIP (Routing Information Protocol) using Cisco Packet Tracer.

Observation :-

1.) Three routers are connected to form a network with multiple possible paths.
2.) IP address are assigned to the PCs, FastEthernet interfaces and Serial interfaces.
3.) RIP is enabled using the router rip command.
4.) Directly connected networks are advertised using the network command.
5.) RIP automatically exchanges routing information between routers.
6.) RIP selects the router based on the lowest hop count.
7.) The ping command is used to test connectivity between PC0 and PC1
8.) The tracert command is used to ~~test connectivity~~ verify the path followed by packets.
9.) If the selected route fails, RIP automatically chooses another available route


PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.0.0.1
Pinging 20.0.0.1 with 32 bytes of data:
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Ping statistics for 20.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Reply from 10.0.0.2: bytes=32 time=18ms TTL=126
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 18ms, Maximum = 19ms, Average = 18ms
C:\>tracert 10.0.0.2
Tracing route to 10.0.0.2 over a maximum of 30 hops:
1 0 ms 0 ms 0 ms 20.0.0.1
2 11 ms 13 ms 12 ms 192.168.1.254
3 9 ms 5 ms 5 ms 10.0.0.2
Trace complete.
C:\>
Top




```
C:\> tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    6 ms      8 ms      7 ms      192.168.1.253
  3    5 ms      10 ms     7 ms      20.0.0.2

Trace complete.
```

Result :- Thus, RIP was successfully configured and dynamic routing was verified. The router automatically selected the available route with the lower hop count and